National 5 Engineering Science

Analogue Electronics Worksheet – Mixed Questions

Show all working. Use correct units.

1. A 12V supply is connected to a 240Ω resistor. Calculate the current.

$I = V / R = 12 / 240 =$ **0.05 A**

2. Two resistors of 15Ω and 35Ω are connected in series to a 10V supply.
a) Calculate the total resistance.

$R_T = 15 + 35 =$ **50Ω**

b) Calculate the circuit current.

$I = 10 / 50 =$ **0.2A**

c) Calculate the voltage across each resistor.

$V_1 = 0.2 \times 15 =$ **3V**
$V_2 = 0.2 \times 35 =$ **7V**

3. Two resistors of 100Ω and 300Ω are connected in parallel to a 12V supply.
a) Calculate the total resistance.

$1/R_T = 1/100 + 1/300 = 0.01 + 0.0033$
$R_T =$ **75Ω**

b) Calculate the total circuit current.

$I_T = 12 / 75 =$ **0.16A**

c) Calculate the current through the 100Ω resistor.

$I_{(100\Omega)} = 12 / 100 =$ **0.12A**

4. A current of 0.5A flows through a resistor connected to a 9V supply. Calculate the resistance.

$R = V / I = 9 / 0.5 =$ **18Ω**

5. A 20Ω resistor is connected in series with two resistors (60Ω and 30Ω) in parallel. The supply voltage is 18V.
a) Calculate the total resistance.

$R_{Parallel} = 20Ω$

$R_T = 20 + 20 =$ **40Ω**

b) Calculate the circuit current.

$I = 18 / 40 =$ **0.45A**

c) Calculate the voltage across the 20Ω resistor.

$V_{(20Ω)} = 0.45 \times 20 =$ **9V**

6. R1 = 2.2kΩ, R2 = 4.7kΩ, Vs = 12V.
a) Calculate total resistance.

$R_T = R_1 + R_2 = 2.2 + 4.7 =$ **6.9kΩ**

b) Calculate circuit current.

$I = 12 / 6900 = 0.$**00174A**

c) Calculate the voltage across R2.

$V_{(4.7kΩ)} =$ **8.17V**

7. Three resistors (10Ω, 20Ω, 40Ω) are connected in parallel to a 12V supply.
a) Calculate total resistance.

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_T} = \frac{1}{10} + \frac{1}{20} + \frac{1}{40} = 0.1 + 0.05 + 0.025 = 0.175$$

R_T = 1 / 0.18333333 = 5.7142857 = **5.71Ω**

b) Calculate total current.

I_T = 12 / 5.71 = **2.1A**

c) Calculate current through the 20Ω resistor.

V = IR

I = V / R

$I_{(20Ω)}$ = 12 / 20 = **0.6A**

8. A current of 2A flows through a 15Ω resistor. Calculate the supply voltage.

V = I × R = 2 × 15 = 30V

9. R1 = 100Ω, R2 = 150Ω, R3 = 250Ω, Vs = 20V.
a) Calculate total resistance.

$R_T = R_1 + R_2 + R_3$ = 100 + 150 + 250 = **500Ω**

b) Calculate circuit current.

I = 20 / 500 = **0.04A**

c) Calculate the voltage across R3.

$V_{(250Ω)}$ = IR = 0.04 x 250 = **10V**

10. Two resistors of 8Ω and 12Ω are connected in parallel to a 6V supply.
a) Calculate total resistance.

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{8 \times 12}{8 + 12}$$

R_T = **4.8Ω**

b) Calculate total current.

I_T = 6 / 4.8 = **1.25A**

11. A 50Ω resistor is connected in series with a parallel pair of 100Ω and 100Ω resistors. The supply voltage is 10V.
a) Calculate total resistance.

$$R_{Parallel} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{100 \times 100}{100 + 100}$$

$R_{Parallel}$ = 50Ω

$R_T = R_3 = R_{parallel}$ = 50 + 50 = **100Ω**

b) Calculate circuit current.

I = 10 / 100 = 0.1A

c) Calculate current through one of the 100Ω resistors.

First calculate voltage across 'in series' resistor:

$V_{50Ω}$ = IR = 0.1 x 50 = 5V

Therefore, voltage across parallel part of circuit is also 5V (5V + 5V = 10V = supply voltage)

Current across entering parallel resistors part of circuit is 0.1A and then it is split.

$I_{100(parallel)} = V_{parallel}$ / R = 5 / 100

$I_{100(branch)}$ = **0.05A**

12. A student needs a 25Ω resistor but only has two 50Ω resistors.
a) Draw a circuit to achieve 25Ω.

Parallel connection drawn

b) Prove your answer with calculations.

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{50 \times 50}{50 + 50}$$

R_T = **25Ω**

13. A 24V supply is connected to a 3kΩ resistor. Calculate the current (in mA).

I = 24 / 3000 = 0.008A = **8mA**

14. Two resistors (1kΩ and 2kΩ) are connected in parallel to a 9V supply.
a) Calculate total resistance.

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{1000 \times 2000}{1000 + 2000}$$

R_T = 666.66667 = **667Ω**

b) Calculate total current.

I_T = 9 / 667 = **0.0135A**

15. Two resistors (30Ω and 60Ω) are connected in parallel. This combination is then connected in series with a 10Ω resistor. Supply voltage is 12V.
a) Total resistance

$$R_{Parallel} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{30 \times 60}{30 + 60}$$

$R_{Parallel}$ = 20Ω

$R_T = R_3 = R_{parallel}$ = 20 + 10 = **30Ω**

b) Circuit current

I = 12 / 30 = **0.4A**

c) Voltage across the 10Ω resistor

First calculate voltage across 'in series' resistor:

$V_{10\Omega}$ = IR = 0.4 x 10 = **4V**

d) Current through the 30Ω resistor

$I_{(30\Omega)}$ = 0.267A
Therefore, voltage across parallel part of circuit is

$V_{total} = V_{10\Omega} + V_{parallel}$

$V_{parallel} = V_{total} - V_{10\Omega}$ = 12 – 4 = 8V (and this is the same voltage in each branch)

Current across entering parallel resistors part of circuit is 0.4A and then it is split.

$I_{30(parallel)} = V_{parallel} / R = 8 / 30$

$I_{30(parallel)}$ = 0.26666 **= 0.27A**

16. R1 = 470Ω, R2 = 1.5kΩ, Vs = 9V.
a) Calculate total resistance.

$R_T = R_1 + R_2$ = 470 + 1500

R_T = **1970Ω**

b) Calculate circuit current.

I = 9 / 1970 = 0.00457A

17. Two heaters of resistance 20Ω and 30Ω are connected in parallel to a 12V battery.
a) Calculate total resistance.

$$R_{total} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{20 \times 30}{20 + 30}$$

R_T = **12Ω**

b) Calculate total current drawn from the battery.

$I_T = 12 / 12 =$ **1A**

c) Calculate current through the 20Ω heater.

$I_{(20\Omega)} = V / R = 12 / 20$

$I_{(20\Omega)} =$ **0.6A**

18. A circuit has a total resistance of 40Ω. One resistor of 10Ω is in series with a parallel combination.
Calculate the resistance of the parallel section.

$R_{Parallel} = 30\Omega$

19. Resistors of 2kΩ, 5kΩ and 10kΩ are connected in parallel to a 10V supply.
a) Calculate total resistance.

$R_T = 1250\Omega$

b) Calculate total current.

$I_T = 10 / 1250 = 0.008A$

c) Calculate current through the 2kΩ resistor.

$I_{(2k\Omega)} = 5mA$

20. A 6V supply feeds a circuit where a 15Ω resistor is in series with two 30Ω resistors connected in parallel.
a) Calculate total resistance.

$R_{parallel} = 15\Omega$
$R_T = 30\Omega$

b) Calculate circuit current.

$I = 6 / 30 = 0.2A$

c) Calculate voltage across the parallel section.

$V_{(parallel)} = 3V$

d) Calculate current through one of the 30Ω resistors.

$I_{(30\Omega)} = 0.1A$